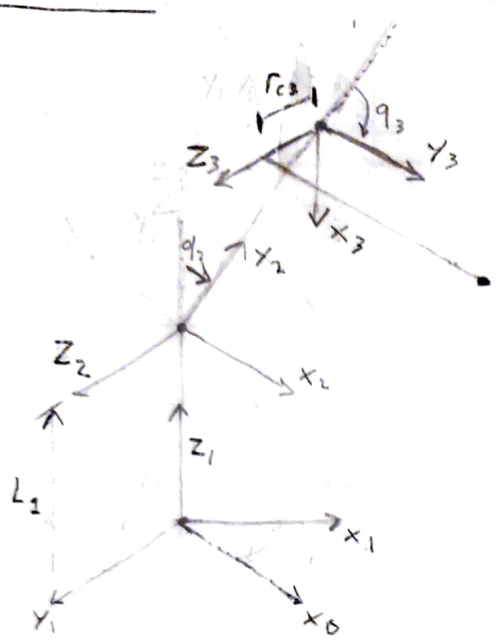
Assignment 1 | ME 599/699 | Daniel Kennedy, Joshua Ashley

Geometry:



Parameters:

$L_1 = 0.4$
$L_{c1} = 0.2$
$L_2 = 1.0$
$L_{c2} = 0.5$
$L_3 = 1.0$
$L_{c3} = 0.5$
$r_{c3} = 0.1$

Transformations:

$$R_1^0 = \begin{bmatrix} c_1 & -s_1 & 0 \\ s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad O_1^0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad T_1^0 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Z_0 rotation by q_1

$$R_{c1}^1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad O_{c1}^1 = \begin{bmatrix} 0 \\ 0 \\ L_{c1} \end{bmatrix} \qquad T_{c1}^1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & L_{c1} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_2^1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} c_2 & -s_2 & 0 \\ s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad O_2^1 = \begin{bmatrix} 0 \\ L_1 \\ 0 \end{bmatrix} \qquad T_2^1 = \begin{bmatrix} c_2 & -s_2 & 0 & 0 \\ 0 & 0 & -1 & L_1 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

X_1 rotation $\qquad$ Z_1 rotation by q_2

$$R_{c2}^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad O_{c2}^2 = \begin{bmatrix} 0 \\ L_{c2} \\ 0 \end{bmatrix} \qquad T_{c2}^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & L_{c2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_3^2 = \begin{bmatrix} c_3 & -s_3 & 0 \\ s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad O_3^2 = \begin{bmatrix} 0 \\ L_2 \\ 0 \end{bmatrix} \qquad T_3^2 = \begin{bmatrix} c_3 & -s_3 & 0 & 0 \\ s_3 & c_3 & 0 & L_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Z_2 rotation by q_3

$$R_{c3}^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad O_{c3}^3 = \begin{bmatrix} 0 \\ L_{c3} \\ r_{c3} \end{bmatrix} \qquad T_{c3}^3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & L_{c3} \\ 0 & 0 & 1 & r_{c3} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Transformations (cont).

$$T^0_{c1} = T^0_1 \, T^1_{c1} = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & L_{c1} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T^0_2 = T^0_1 \, T^1_2 = \begin{bmatrix} c_1 c_2 & -c_1 s_2 & s_1 & -L_1 s_1 \\ s_1 c_2 & -s_1 s_2 & -c_1 & L_1 c_1 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T^0_{c2} = T^0_2 \, T^2_{c2} = \begin{bmatrix} c_1 c_2 & -c_1 s_2 & s_1 & -L_1 s_1 - L_{c2} c_1 s_2 \\ s_1 c_2 & -s_1 s_2 & -c_1 & c_1 L_1 - L_{c2} s_1 s_2 \\ s_2 & c_2 & 0 & L_{c2} c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T^0_3 = T^0_2 \, T^2_3 = \begin{bmatrix} c_1 c_2 c_3 - c_1 s_2 s_3 & -c_1 c_2 s_3 - c_1 s_2 c_3 & s_1 & -L_1 s_1 - L_2 c_1 s_2 \\ s_1 c_2 c_3 - s_1 s_2 s_3 & -s_1 c_2 s_3 - s_1 s_2 c_3 & -c_1 & L_1 c_1 - L_2 s_1 s_2 \\ c_2 s_3 - s_2 c_3 & c_2 c_3 - s_2 s_3 & 0 & L_2 c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T^0_{c3} = T^0_3 \, T^3_{c3} = \begin{bmatrix} c_1 c_2 c_3 - c_1 s_2 s_3 & -c_1 c_2 s_3 - c_1 s_2 c_3 & s_1 & -L_1 s_1 - c_1 L_2 s_2 - L_{c3} c_1 c_2 s_3 - L_{c3} c_1 s_2 c_3 + r_{c3} s_1 \\ s_1 c_2 c_3 - s_1 s_2 s_3 & -s_1 c_2 s_3 - s_1 s_2 c_3 & -c_1 & L_1 c_1 - r_{c3} c_1 - L_{c3} s_1 c_2 s_3 - L_{c3} s_1 s_2 c_3 - L_2 s_1 s_2 \\ c_2 s_3 - s_2 c_3 & c_2 c_3 - s_2 s_3 & 0 & L_2 c_2 + L_{c3} c_2 c_3 - L_{c3} s_2 s_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_1^0 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad T_{c1}^1 = \begin{bmatrix} I & \begin{bmatrix} 0 \\ 0 \\ L_{c1} \end{bmatrix} \\ 0 & 1 \end{bmatrix}$$

$$T_2^1 = Trans_{z_1}, Rot_{x_2, \pi/2}, Rot_{z_2, \theta_2}$$

$$= \begin{bmatrix} I & \begin{matrix} 0 \\ 0 \\ L_1 \end{matrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 0 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{c2}^2 = \begin{bmatrix} I & \begin{bmatrix} 0 \\ L_{c2} \\ 0 \end{bmatrix} \\ 0 & 1 \end{bmatrix}$$

$$T_3^2 = Trans_{y_2, L_2}, Rot_{z_3, \theta_3}$$

$$= \begin{bmatrix} I & \begin{matrix} 0 \\ L_2 \\ 0 \end{matrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 0 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{c3}^3 = \begin{bmatrix} I & \begin{bmatrix} 0 \\ L_{c3} \\ r_{c3} \end{bmatrix} \\ 0 & 1 \end{bmatrix}$$

$L_{c1} = 0.2$

$L_{c2} = 0.5$

$L_{c3} = 0.5$

$L_1 = 0.4$

$L_2 = 1.0$

$r_{c3} = 0.1$

$$J_{c1} = \begin{bmatrix} R_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_{c1}^0 - O_1^0) \\ R_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix}$$

$$J_{c2} = \begin{bmatrix} R_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_{c2}^0 - O_1^0) & R_2^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_{c2}^0 - O_2^0) \\ R_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & R_2^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix}$$

$$J_{c3} = \begin{bmatrix} R_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_{c3}^0 - O_1^0); & R_2^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_{c3}^0 - O_2^0); & R_3^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_{c3}^0 - O_3^0) \\ R_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & ; R_2^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & ; R_3^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix}$$

Jacobians:

$$J_i = \begin{bmatrix} R_{i-1}^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_n^0 - O_{i-1}^0) \\ R_{i-1}^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix}$$

$$J_{c1,v}^0 = \begin{bmatrix} R_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_{c1}^0 - O_1^0) & \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$J_{c1,\omega}^0 = \begin{bmatrix} R_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{bmatrix}$$

$$J_{c2,v}^0 = \begin{bmatrix} R_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_{c2}^0 - O_1^0) & R_2^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_{c2}^0 - O_2^0) & \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{bmatrix}$$

$$J_{c2,\omega}^0 = \begin{bmatrix} R_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & R_2^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{bmatrix}$$

$$J_{c3,v}^0 = \begin{bmatrix} R_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_{c3}^0 - O_1^0) & R_2^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_{c3}^0 - O_2^0) & R_3^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times (O_{c3}^0 - O_3^0) \end{bmatrix}$$

$$J_{c3,\omega}^0 = \begin{bmatrix} R_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & R_2^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & R_3^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix}$$

Local Inertia Tensors:

$$I_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.083 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad I_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.083 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.33 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$m_1 = 1$ $\qquad$ $m_2 = 1$ $\qquad$ $m_3 = 1$

Kinetic Energy

$$KE = \sum_{i=1}^{3} \left(\frac{1}{2} \dot{q}^T J_{c_i,\omega}^T \left(R_i^0 I_i R_i^{0T} \right) J_{c_i,\omega} \right.$$

(rotational kinetic energy)

$$\left. + m_i J_{c_i,v}^T J_{c_i,v} \right)$$

← closed form soln output from MATLAB code (attached)

(translational kinetic energy)

Potential Energy

$$PE = \sum_{i=1}^{3} \left(m_i g \left(O_{c_i}^0 \cdot [0\ 0\ 1] \right) \right)$$

(z component of c_i)

Lagrangian

$$\mathcal{L} = KE - PE = \mathcal{L}(q, \dot{q})$$

Euler-Lagrangian Dynamics

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = \tau_i$$

← Also provided by MATLAB function